

Does Cold Weather Make Arthritis Worse?

Explore the link between cold weather and arthritis pain. Evidence-based tips for managing symptoms during winter months.

By **Living With Arthritis Editorial Team** · Reviewed by **Clinical Advisory Panel** (HCPC-registered physiotherapy & rheumatology nursing input) · 2026-03-08 · livingwitharthritis.org.uk

The Cold-Arthritis Connection



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Nearly 70% of people with arthritis report that cold weather worsens their symptoms. But does the science support this widespread belief?

What the Research Says

The relationship between weather and arthritis pain is complex:

- A 2019 study of 13,000 UK participants (published in *npj Digital Medicine*) found a ****significant association**** between humidity, wind speed, and joint pain — but temperature alone was not the strongest predictor
- Barometric pressure drops (common before storms) may cause joint tissues to expand slightly, irritating nerve endings
- Cold temperatures reduce blood flow to extremities, increasing stiffness
- Reduced activity in winter leads to deconditioning, worsening symptoms



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Why Joints Hurt More in Winter

Physiological Factors - **Synovial fluid thickens** in cold temperatures, reducing joint lubrication - **Barometric pressure changes** affect joint capsule pressure - **Muscle tension increases** as the body tries to conserve heat - **Vitamin D drops** — the UK's latitude means no useful UV from October to March

Behavioural Factors - Less physical activity and outdoor exercise - Heavier clothing restricting movement - Lower mood affecting pain perception - Comfort eating replacing anti-inflammatory diets

Winter-Proofing Your Joints



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Keep Warm - Layer clothing, especially over affected joints - Use thermal gloves and knee warmers - Warm up indoors before going outside - Consider heated blankets or wheat bags

Stay Active - Indoor swimming pools maintain warm water (28–32°C) - Home exercise routines — no weather excuse needed - Indoor cycling or stationary bike - Yoga and tai chi classes (many available online)

Dietary Support - Increase omega-3 intake (oily fish, walnuts, flaxseed) - Supplement vitamin D (10 mcg/day minimum, October–March) - Warm anti-inflammatory drinks: turmeric latte, ginger tea - Hearty Mediterranean-style soups and stews

Environment - Keep your home warm (at least 18°C) - Use dehumidifiers to reduce dampness - Warm your car before driving - Consider a heated steering wheel cover

When It Is More Than the Weather

If your symptoms are significantly worse in winter, speak to your GP — it may indicate: - Inadequately managed inflammation - Need for medication adjustment - Seasonal Affective Disorder compounding pain

Reviewed by Dr. Michael Chen, Consultant Rheumatologist, Guy's and St Thomas' NHS Foundation Trust

Q&A

Question: What is the key thing to know about does cold weather make arthritis worse??

Short answer: Cold weather can worsen arthritis symptoms for many people, although temperature alone isn't the only factor. Humidity, wind speed, and drops in barometric pressure can all play a role. Physiologically, cold can thicken synovial fluid, reduce blood flow, and increase muscle tension. Reduced activity and lower vitamin D levels in winter can also contribute to increased joint pain.

Question: When should I seek professional advice?

Short answer: If symptoms persist or worsen, pain is severe and unresponsive to usual relief, or you develop new symptoms, contact your GP, physiotherapist or rheumatology team promptly.

Question: Where can I get free arthritis support in the UK?

Short answer: Arthritis UK: 0800 5200 520. NRAS: 0800 298 7650. Free tools and guides at livingwitharthritis.org.uk.

General information only — not medical advice. Consult your GP or specialist before changing medication or starting a new programme. Emergency: 999. NHS advice: 111. Living With Arthritis · livingwitharthritis.org.uk